

Tim Keller, PhD

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For more details, particularly regarding projects, please refer to 🔗 <https://timkeller15.github.io>.

PROFILE

Theoretical physicist with expertise in quantum optics, ultracold atomic gases, and high-performance computing for numerical simulations. Experienced in developing analytical models, optimal control strategies, and custom GPU-accelerated software for a broad variety of quantum systems from individual atoms to many-body cavity QED setups.

Proven ability to independently design and execute complex research projects from theoretical modeling and large-scale simulation to data analysis and publication in peer-reviewed journals, while collaborating effectively across theoretical and experimental teams.

EXPERIENCE

- **Junior Researcher** **since Apr 2023**
Waseda University, Tokyo, Japan
 - funded by JST Moonshot R&D Goal 6 Project: *Large-scale quantum hardware based on nanofiber cavity QED* led by 🔗 Prof. Takao Aoki
 - developed theoretical models and numerical simulations to apply quantum computing performance benchmarks to neutral-atom qubits in a nanofiber cavity QED platform, guiding experimental optimization of quantum gate fidelities
 - designed a custom GPU-accelerated numerical FDTD solver for Maxwell's equations to model laser-induced trapping potentials near optical nanofibers and calculate fast and efficient atom-loading protocols
- **JSPS Postdoctoral Fellow** **Sep 2022 - Mar 2023**
OIST, Okinawa, Japan
 - extended PhD work on many-body quantum dynamics and control of interacting quantum gases
 - supervised and mentored three PhD students and two research interns, providing training in numerical simulation and analytical techniques

EDUCATION

- **PhD in Physics** **Sep 2017 - Aug 2022**
Thesis: 📄 Controlling Superfluid and Insulating States in Interacting Quantum Gases
Okinawa Institute of Science and Technology (OIST), Japan **Advisor:** Prof. Thomas Busch
- **MSc in Physics** **Oct 2014 - Dec 2016**
Saarland University, Germany **Advisor:** Prof. Giovanna Morigi
- **BSc in Physics** **Oct 2011 - Aug 2014**
Saarland University, Germany **Advisor:** Prof. Giovanna Morigi

FELLOWSHIPS & AWARDS

- **Research Fellowship for Young Scientists (DC2)** **Apr 2021 - Mar 2023**
Japan Society for the Promotion of Science **Grant:** 🔗 JSPS KAKENHI 21J10521
Stipend: JPY 200 000 (~ USD 1800) per month & Research Grant of JPY 900 000 (~ USD 8200)

- **Outstanding Student Award** **Apr 2021 - Mar 2023**
Okinawa Institute of Science and Technology Graduate University
 Annual tuition of JPY 540 000 (~ USD 4800) waived in recognition of JSPS fellowship
- **Physical Review Letters Editor's Suggestion** **Jan 2022**
American Physical Society **Publication:**  PRL **128**, 053401 (2022).
 Flagship journal with acceptance rate < 25%. Only one in seven Letters highlighted as Editor's Suggestion due to their particular importance, innovation, and broad appeal.

PROJECTS & PUBLICATIONS

- (9) **Hikari.jl - GPU-accelerated FDTD Maxwell equation solver for calculating laser-generated atomic trapping potentials close to optical nanofibers**
 Contribution: *T. Keller* Code:  Hikari.jl
- (8) **Addressing requirements for crosstalk-free quantum-gate operation in many-body nanofiber cavity QED systems**
 Contribution: *T. Keller*, *S. Kikura*, *R. Asaoka*, *Y. Suzuki*, *Y. Tokunaga*, and *T. Aoki*
 Publication:  arXiv:2509.08408 (2025). Code:  AddressingRequirements.jl
- (7) **Fermionization of a few-body Bose system immersed into a Bose-Einstein condensate**
 Contribution: *T. Keller*, *T. Fogarty*, and *T. Busch*
 Publication:  SciPost Phys. **15**, 095 (2023). Code:  Fermionization
- (6) **Self-Pinning Transition of a Tonks-Girardeau Gas in a Bose-Einstein Condensate**
 Contribution: *T. Keller*, *T. Fogarty*, and *T. Busch*
 Publication:  Phys. Rev. Lett. **128**, 053401 (2022). (Editor's Suggestion) Code:  Self-Pinning
- (5) **Adiabatic critical quantum metrology cannot reach the Heisenberg limit even when shortcuts to adiabaticity are applied**
 Contribution: *K. Gietka*, *F. Metz*, *T. Keller*, and *J. Li*
 Publication:  Quantum **5**, 489 (2021).
- (4) **Feshbach engine in the Thomas-Fermi regime**
 Contribution: *T. Keller*, *T. Fogarty*, *J. Li*, and *T. Busch*
 Publication:  Phys. Rev. Research **2**, 033335 (2020). Code:  Feshbach-Engine
- (3) **Quenches across the self-organization transition in multimode cavities**
 Contribution: *T. Keller*, *V. Torggler*, *S. Jäger*, *S. Schütz*, *H. Ritsch*, and *G. Morigi*
 Publication:  New J. Phys. **20** 025004 (2018). Code:  MultimodeSelforganization.jl
- (2) **Phases of cold atoms interacting via photon-mediated long-range forces**
 Contribution: *T. Keller*, *S. Jäger*, and *G. Morigi*
 Publication:  J. Stat. Mech. 064002 (2017). Code:  MultimodeSelforganization.jl
- (1) **Probing the out-of-equilibrium dynamics of two interacting atoms**
 Contribution: *T. Keller* and *T. Fogarty*
 Publication:  Phys. Rev. A **94**, 063620 (2016). Code:  TwoAtomQuenches.jl

LANGUAGES

German: Native | **English:** Fluent (TOEFL Score 114/120) | **Japanese:** Intermediate (JLPT N2 Certificate)